THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME IV

The Circular-Polarisation Decomposition
and the Birth of the Transverse Magnetisation:
Equations (SM-16)–(SM-24)

Where Volume III ended. Volume III assembled the complete ferrohydrodynamic system (SM-6)–(SM-10), eliminated the particle spin to obtain the reduced pair (SM-12)–(SM-13), specified the applied field $\boldsymbol{H} = (H_0 \cos \omega t, 0, 0)$ in (SM-14), and adopted the linear magnetisation law $\boldsymbol{M}_0 = \chi \boldsymbol{H}$, $\chi = nm^2/3kT$, in (SM-15). It ended by quoting the first line of the solution to come.

Volume IV carries out that solution. The strategy of Shliomis & Morozov's "first variant" is to write the linearly polarised field as the sum of two counter-rotating circular fields, solve for each separately — a problem in which nothing depends on time in the co-rotating frame — and then superpose. The payoff is Eq. (SM-23): a transverse magnetisation *proportional to the vorticity*, which is the object that will produce the torque, the slip and finally the negative viscosity. Chapter numbers resume at 30, page numbers at 113.

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Part IX

The Circular-Polarisation Solution

Chapter 30

Equation (SM-16): Decomposition into Two Rotating Fields

$$\boldsymbol{H}_{\pm} = \big(H_0 \cos\omega t,\ \pm H_0 \sin\omega t,\ 0\big), \qquad \boldsymbol{H} = \tfrac{1}{2}\big(\boldsymbol{H}_{+} + \boldsymbol{H}_{-}\big). \tag{SM-16}$$

30.1 Context

The authors introduce their Sec. II A with the words: "In the first the linear-polarized field (14) is presented as a sum of two fields, which are circularly polarized opposite each other. This approach is rather visual and in our opinion it makes clear the physical nature of the problem." That is exactly right, and it is why we follow it before the shorter formal route of Sec. II B (which will be Volume V, Chapter 44).

The reason the decomposition helps is this. In the linearly polarised field of (SM-14) *everything* depends on time: the field grows, dies, reverses. In a circularly polarised field nothing does — the field merely rotates, and in a frame turning with it the problem is completely static. A static problem is one we can solve exactly. Superposing the two solutions then reconstructs the oscillating case.

30.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{H}_{+}$	left-hand (counterclockwise) circularly polarised field, rotating from $+x$ towards $+y$	vector field	Oe
$\boldsymbol{H}_{-}$	right-hand (clockwise) circularly polarised field	vector field	Oe
H_0	amplitude of each rotating field, equal to the amplitude of the linear field	scalar	Oe
ω	angular frequency of rotation	scalar	s^{-1}

Note that $|\boldsymbol{H}_{\pm}| = H_0$ at *all* times:

$$|\boldsymbol{H}_{\pm}|^2 = H_0^2 \cos^2\omega t + H_0^2 \sin^2\omega t = H_0^2, \tag{30.2.1}$$

by the Pythagorean identity. A circularly polarised field has constant magnitude and rotating direction; a linearly polarised one has fixed direction and oscillating magnitude. They are

opposite in every respect, which is why the decomposition is illuminating rather than merely algebraic.

30.3 Assumptions

None beyond those of (SM-14): spatial uniformity and harmonic time dependence. The decomposition itself is an identity, not an approximation.

Remark 30.3.1 (But the *use* of the decomposition does assume linearity). Splitting the field is exact. Splitting the *response* — writing $\boldsymbol{M} = \frac{1}{2}(\boldsymbol{M}_+ + \boldsymbol{M}_-)$ — requires that the magnetisation depend linearly on the field, i.e. Assumption 32.3.1 of Chapter 32. The authors return to this point explicitly beneath their Eq. (SM-24), and we discuss it in Sec. 37.6. Keep it in mind: it is the single subtlety of the whole chapter.

30.4 Verification of the Identity

Step 1: add the two fields componentwise.

$$\boldsymbol{H}_+ + \boldsymbol{H}_- = \big(H_0\cos\omega t,\ +H_0\sin\omega t,\ 0\big) + \big(H_0\cos\omega t,\ -H_0\sin\omega t,\ 0\big) \tag{30.4.1}$$

$$= \big(H_0\cos\omega t + H_0\cos\omega t,\ \ H_0\sin\omega t - H_0\sin\omega t,\ \ 0\big) \tag{30.4.2}$$

$$= \big(2H_0\cos\omega t,\ 0,\ 0\big). \tag{30.4.3}$$

Step 2: halve.

$$\tfrac{1}{2}\big(\boldsymbol{H}_+ + \boldsymbol{H}_-\big) = \big(H_0\cos\omega t,\ 0,\ 0\big) = \boldsymbol{H}, \tag{30.4.4}$$

which is exactly the field (SM-14). ■

30.5 Physical Interpretation

Picture two vectors of equal length H_0, both starting along $+x$ at $t = 0$, one turning counterclockwise and one clockwise at the same rate ω. At every instant their y components are equal and opposite and cancel, while their x components are equal and add. The sum therefore stays on the x axis and its length oscillates as $2H_0\cos\omega t$. This is the standard fact that a linear oscillation is the superposition of two opposite circular motions — the same statement that in optics decomposes plane-polarised light into left- and right-circular components.

Why this is the physically right decomposition here. The suspended grains care about *rotation*. A circularly polarised field drags the moments around continuously and therefore drives a steady, unidirectional particle spin. The + field drives spin along $+z$; the − field drives spin along $-z$. In the linearly polarised sum the two drives are exactly equal and opposite — which is precisely the statement, made qualitatively in Sec. 27.5, that half the grains swing one way and half the other, so that $\boldsymbol{\Theta} = 0$ in a quiescent fluid. What the decomposition buys us is a way to see *quantitatively* how a shear unbalances the two.

30.6 Connection to the Next Equation

We must now solve the ferrohydrodynamic system for one circular field. The form of that solution is (SM-17).

Chapter 31

Equation (SM-17): The Steadily Rotating Solution

$$\boldsymbol{M}_{\pm} = \big[M\cos(\omega t - \alpha),\ \pm M\sin(\omega t - \alpha),\ 0\big], \qquad \boldsymbol{\Theta}_{\pm} = \big(0, 0, \pm\Theta\big). \tag{SM-17}$$

31.1 Context

This is the *ansatz*: the form of the solution, to be justified now and verified by substitution in Chapter 32. The authors say: "For each of these fields taken separately, Eqs. (6)–(8) admit a solution, in which the fluid is quiescent ($\boldsymbol{\Omega} = 0$) and the magnetization rotates with an angular frequency of the field, lagging behind it at some angle α. The ferroparticles rotate with a stationary angular velocity, which is oriented perpendicularly to the plane of the field rotation."

31.2 Symbols

Symbol	Meaning	Type	Units
M	magnitude of the rotating magnetisation, constant in time	scalar	G
α	phase lag of the magnetisation behind the field	scalar, $0 \le \alpha < \pi/2$	rad
Θ	magnitude of the steady particle spin	scalar	s^{-1}
$\omega t - \alpha$	instantaneous angle of $\boldsymbol{M}$ in the xy plane	scalar	rad

31.3 Assumptions

1. **The fluid is at rest**, $\boldsymbol{\Omega} = 0$. This chapter deals with the unperturbed problem; the vortex is switched on in Chapter 34.

2. **Steady state in the rotating frame.** All transients have decayed; this requires only that we wait a few τ_B after switching on, which is ∼ms.

3. **The field amplitude is weak**, so that $\boldsymbol{M}_0 = \chi\boldsymbol{H}$ (SM-15) holds and the response is linear.

31.4 Why the Solution Must Have This Form

The argument is a symmetry argument and is worth spelling out, because it is what makes the ansatz inevitable rather than lucky.

Step 1: the problem has a symmetry. Consider the + field. Rotate the entire problem about the z axis by an angle δ and simultaneously shift time by δ/ω. The field is unchanged:

$$\boldsymbol{H}_+\big(t+\delta/\omega\big) = H_0\big(\cos(\omega t+\delta),\,\sin(\omega t+\delta),\,0\big) = \mathsf{R}_z(\delta)\,\boldsymbol{H}_+(t), \tag{31.4.1}$$

where $\mathsf{R}_z(\delta)$ is the rotation matrix. All the other ingredients of the problem — the isotropic carrier liquid, the spherical grains, the temperature — are unchanged by a rotation. So the problem is invariant under the combined operation.

Step 2: therefore the solution is too. A unique steady solution of an invariant problem must be invariant. Hence $\boldsymbol{M}(t+\delta/\omega) = \mathsf{R}_z(\delta)\boldsymbol{M}(t)$ for every δ, which says precisely that $\boldsymbol{M}$ rotates rigidly in the xy plane at angular velocity ω with a fixed magnitude. Writing its angle as $\omega t-\alpha$ gives the first line of (SM-17). Because $\boldsymbol{H}$ began on the x axis, α is the amount by which $\boldsymbol{M}$ trails it.

Step 3: the direction of $\boldsymbol{\Theta}$. Likewise $\boldsymbol{\Theta}$ must be invariant under R_z, and the only vectors invariant under all rotations about z are those along z. Hence $\boldsymbol{\Theta} = (0,0,\Theta)$, with Θ a constant. This is the paper's "oriented perpendicularly to the plane of the field rotation".

Step 4: the sign for the two fields. The − field is the mirror image of the + field in the xz plane ($y\to -y$). The magnetisation, being a sum of moments carried by the grains, transforms the same way, giving the $\pm$ in front of $M\sin(\omega t-\alpha)$. The spin $\boldsymbol{\Theta}$ is a pseudo-vector along z and reverses under this mirror, giving $\pm\Theta$. The magnitudes M, α, Θ are the *same* for both fields — a fact we shall use in Chapter 33 and lose in Chapter 34, where the vortex destroys the mirror symmetry.

31.5 Physical Interpretation of the Lag

Why must $\boldsymbol{M}$ lag rather than lead? Because relaxation is dissipative. The magnetisation is always trying to reach $\boldsymbol{M}_0$, which points along the present field; but reaching it takes a time τ_B, during which the field has moved on by $\omega\tau_B$ radians. The magnetisation is therefore permanently behind by an angle of that order, and we shall find in Chapter 32 that the relation is exactly $\tan\alpha = \omega\tau_B$ when the back-reaction is neglected.

This lag is the whole story: a magnetisation parallel to the field exerts no torque, by (1.5.7). Only because $\alpha\neq 0$ is there a torque $\boldsymbol{M}\times\boldsymbol{H}$ at all — and that torque is what spins the particles.

31.6 Connection to the Next Equation

Substituting the ansatz into the equations of motion turns three vector equations into three scalar relations among M, α and Θ. Those are (SM-18).

Chapter 32

Equation (SM-18): The Three Relations

$$M = \chi H_0 \cos\alpha, \qquad \Theta = \big(\tau_s/I\big) M H_0 \sin\alpha, \qquad \tan\alpha = \big(\omega - \Theta\big)\tau_B. \tag{SM-18}$$

32.1 Context

The paper says simply: "Substituting (15)–(17) into Eqs. (7) and (8), we obtain (18)." We now perform that substitution completely.

32.2 Symbols

All defined. $\chi = nm^2/3kT$ is the susceptibility (SM-15); $\tau_s/I = 1/6\eta\varphi$ by Eq. (7.5.12).

32.3 Assumptions

Assumption 32.3.1 (Weak field)**.** $\xi_0 = mH_0/kT \ll 1$, so that $\boldsymbol{M}_0 = \chi \boldsymbol{H}$ (SM-15). Without this the Langevin nonlinearity would forbid the superposition of Chapter 37.

Plus Assumptions 1–2 of Chapter 31 (quiescent fluid, steady state).

32.4 Mathematical Background: the co-rotating basis

Because everything rotates at the rate ω, it pays to introduce two unit vectors that rotate *with the magnetisation*. Write

$$\phi \equiv \omega t - \alpha \tag{32.4.1}$$

for the instantaneous angle of $\boldsymbol{M}$, and define

$$\hat{\boldsymbol{e}}_r \equiv \big(\cos\phi,\ \sin\phi,\ 0\big), \qquad \hat{\boldsymbol{e}}_t \equiv \big(-\sin\phi,\ \cos\phi,\ 0\big). \tag{32.4.2}$$

The subscripts stand for "radial" (along $\boldsymbol{M}$) and "tangential" (perpendicular to it, in the direction of increasing ϕ).

Property 1: they are orthonormal.

$$\hat{\boldsymbol{e}}_r \cdot \hat{\boldsymbol{e}}_r = \cos^2\phi + \sin^2\phi = 1, \tag{32.4.3}$$

$$\hat{\boldsymbol{e}}_t \cdot \hat{\boldsymbol{e}}_t = \sin^2\phi + \cos^2\phi = 1, \tag{32.4.4}$$

$$\hat{\boldsymbol{e}}_r \cdot \hat{\boldsymbol{e}}_t = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0. \tag{32.4.5}$$

Property 2: their time derivatives. Since $\mathrm{d}\phi/\mathrm{d}t = \omega$,

$$\frac{\mathrm{d}\hat{\boldsymbol{e}}_r}{\mathrm{d}t} = \big(-\sin\phi,\ \cos\phi,\ 0\big)\,\omega = \omega\,\hat{\boldsymbol{e}}_t, \tag{32.4.6}$$

$$\frac{\mathrm{d}\hat{\boldsymbol{e}}_t}{\mathrm{d}t} = \big(-\cos\phi,\ -\sin\phi,\ 0\big)\,\omega = -\omega\,\hat{\boldsymbol{e}}_r. \tag{32.4.7}$$

Property 3: cross products with $\hat{\boldsymbol{z}}$.

$$\hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_r = \hat{\boldsymbol{e}}_t, \qquad \hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_t = -\hat{\boldsymbol{e}}_r, \qquad \hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_t = \hat{\boldsymbol{z}}. \tag{32.4.8}$$

The first follows from $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -\hat{\boldsymbol{x}}$: $\hat{\boldsymbol{z}} \times (\cos\phi, \sin\phi, 0) = (-\sin\phi, \cos\phi, 0) = \hat{\boldsymbol{e}}_t$. The others are obtained the same way.

Property 4: the field in this basis. The + field is $\boldsymbol{H}_+ = H_0(\cos\omega t, \sin\omega t, 0)$, i.e. a vector of length H_0 at angle $\omega t = \phi + \alpha$. Using the addition formulas $\cos(\phi+\alpha) = \cos\phi\cos\alpha - \sin\phi\sin\alpha$ and $\sin(\phi+\alpha) = \sin\phi\cos\alpha + \cos\phi\sin\alpha$,

$$\boldsymbol{H}_+ = H_0\big(\cos(\phi+\alpha),\ \sin(\phi+\alpha),\ 0\big) \tag{32.4.9}$$

$$= H_0\Big[\cos\alpha\,\big(\cos\phi, \sin\phi, 0\big) + \sin\alpha\,\big(-\sin\phi, \cos\phi, 0\big)\Big] \tag{32.4.10}$$

$$= H_0\big(\cos\alpha\,\hat{\boldsymbol{e}}_r + \sin\alpha\,\hat{\boldsymbol{e}}_t\big). \tag{32.4.11}$$

The geometric content: the field is ahead of the magnetisation by the angle α, so it has a component $H_0\cos\alpha$ along $\boldsymbol{M}$ and a component $H_0\sin\alpha$ perpendicular to it, on the leading side.

32.5 Step-by-Step Derivation

We work with the + field throughout; the − case follows by the mirror symmetry of Sec. 31.4, Step 4.

Step 1: write the ansatz in the rotating basis. By (SM-17) and (32.4.2),

$$\boldsymbol{M} = M\,\hat{\boldsymbol{e}}_r, \qquad \boldsymbol{\Theta} = \Theta\,\hat{\boldsymbol{z}}. \tag{32.5.1}$$

Step 2: the left-hand side of (SM-7). M is constant, so only the basis vector varies:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = M\frac{\mathrm{d}\hat{\boldsymbol{e}}_r}{\mathrm{d}t} = M\omega\,\hat{\boldsymbol{e}}_t, \tag{32.5.2}$$

using (32.4.6).

Step 3: the convective term. With (32.4.8),

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \Theta M\big(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{e}}_r\big) = \Theta M\, \hat{\boldsymbol{e}}_t. \tag{32.5.3}$$

Step 4: the relaxation term. By (SM-15), $\boldsymbol{M}_0 = \chi \boldsymbol{H}_+$, so with (32.4.11)

$$-\frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big) = -\frac{1}{\tau_B}\Big[M\hat{\boldsymbol{e}}_r - \chi H_0 \cos\alpha\, \hat{\boldsymbol{e}}_r - \chi H_0 \sin\alpha\, \hat{\boldsymbol{e}}_t\Big]. \tag{32.5.4}$$

Step 5: assemble (SM-7). Setting (32.5.2) equal to the sum of (32.5.3) and (32.5.4):

$$M\omega\, \hat{\boldsymbol{e}}_t = \Theta M\, \hat{\boldsymbol{e}}_t - \frac{M - \chi H_0 \cos\alpha}{\tau_B}\, \hat{\boldsymbol{e}}_r + \frac{\chi H_0 \sin\alpha}{\tau_B}\, \hat{\boldsymbol{e}}_t. \tag{32.5.5}$$

Step 6: separate the components. Because $\hat{\boldsymbol{e}}_r$ and $\hat{\boldsymbol{e}}_t$ are independent (orthonormal, Property 1), the coefficients of each must match separately. This is the crucial move and it is legitimate exactly because the basis is orthonormal: dotting (32.5.5) with $\hat{\boldsymbol{e}}_r$ and with $\hat{\boldsymbol{e}}_t$ gives two independent scalar equations.

Radial ($\hat{\boldsymbol{e}}_r$) component:

$$0 = -\frac{M - \chi H_0 \cos\alpha}{\tau_B} \qquad \Longrightarrow \qquad \boxed{M = \chi H_0 \cos\alpha} \tag{32.5.6}$$

which is the first relation of (SM-18). ✓

Tangential ($\hat{\boldsymbol{e}}_t$) component:

$$M\omega = \Theta M + \frac{\chi H_0 \sin\alpha}{\tau_B}. \tag{32.5.7}$$

Step 7: extract the phase relation. Substitute $\chi H_0 = M/\cos\alpha$ from (32.5.6) into (32.5.7):

$$M\omega = \Theta M + \frac{M}{\tau_B}\frac{\sin\alpha}{\cos\alpha} \tag{32.5.8}$$

$$= \Theta M + \frac{M \tan\alpha}{\tau_B}. \tag{32.5.9}$$

Divide throughout by M (nonzero) and rearrange:

$$\omega - \Theta = \frac{\tan\alpha}{\tau_B}, \tag{32.5.10}$$

$$\boxed{\tan\alpha = \big(\omega - \Theta\big)\tau_B} \tag{32.5.11}$$

which is the third relation of (SM-18). ✓

Step 8: the particle spin, from (SM-11). With the fluid at rest, $\boldsymbol{\Omega} = 0$, Eq. (SM-11) gives $\boldsymbol{\Theta} = (\tau_s/I)(\boldsymbol{M} \times \boldsymbol{H})$. Compute the torque with (32.5.1), (32.4.11) and (32.4.8):

$$\boldsymbol{M} \times \boldsymbol{H}_+ = M\hat{\boldsymbol{e}}_r \times H_0\big(\cos\alpha\, \hat{\boldsymbol{e}}_r + \sin\alpha\, \hat{\boldsymbol{e}}_t\big) \tag{32.5.12}$$

$$= MH_0 \cos\alpha\, \big(\hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_r\big) + MH_0 \sin\alpha\, \big(\hat{\boldsymbol{e}}_r \times \hat{\boldsymbol{e}}_t\big) \tag{32.5.13}$$

$$= \mathbf{0} + MH_0 \sin\alpha\, \hat{\boldsymbol{z}} \tag{32.5.14}$$

$$= MH_0 \sin\alpha\, \hat{\boldsymbol{z}}. \tag{32.5.15}$$

Hence

$$\boldsymbol{\Theta} = \frac{\tau_s}{I} MH_0 \sin\alpha\, \hat{\boldsymbol{z}} \qquad \Longrightarrow \qquad \boxed{\Theta = \frac{\tau_s}{I} MH_0 \sin\alpha} \tag{32.5.16}$$

the second relation of (SM-18). ✓ ■

32.6 Physical Interpretation

The three relations of (SM-18) form a small self-consistent loop, and it is worth tracing it round once.

- (32.5.6): the magnetisation is smaller than the equilibrium value χH_0 by the factor $\cos\alpha$. *Why:* only the component of the driving field along $\boldsymbol{M}$ can sustain it; the perpendicular component is spent turning it. The bigger the lag, the weaker the magnetisation.
- (32.5.15): the torque is $MH_0 \sin\alpha$, proportional to the *perpendicular* component of the field. It vanishes if $\alpha = 0$ (perfect following, no dissipation) and is maximal at $\alpha = \pi/2$.
- (32.5.16): that torque, divided by the friction, is the spin.
- (32.5.11): and the spin feeds back on the lag. The particle, by turning, carries the magnetisation partway around, so the magnetisation only has to relax through the *relative* rotation $\omega - \Theta$, not the full ω. Hence $\tan\alpha = (\omega - \Theta)\tau_B$, not $\omega\tau_B$.

Remark 32.6.1 (The physical meaning of $\omega - \Theta$). This difference is the frequency of the field *as seen by a grain*. If a grain could be dragged around at exactly the field frequency, $\Theta = \omega$, it would see a static field, there would be no lag, no torque — and hence, by (32.5.16), no spin, contradicting the assumption. The self-consistent compromise is $0 < \Theta < \omega$, and we shall see in Chapter 33 that Θ is in fact smaller than ω by the factor $\xi^2/3$: for weak fields the grains are barely dragged at all.

32.7 Connection to the Next Equation

The three relations of (SM-18) still contain the auxiliary unknown α. The paper now eliminates it, obtaining explicit expressions for M, Θ and $\tan\alpha$ in terms of $\omega\tau_B$ and the field strength. That is (SM-19).

Chapter 33

Equation (SM-19): Eliminating the Phase Angle

$$M = \frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}, \qquad \Theta = \omega\,\frac{\xi^2/3}{1+\omega^2\tau_B^2}, \qquad \tan\alpha = \omega\tau_B\left(1 - \frac{\xi^2/3}{1+\omega^2\tau_B^2}\right). \tag{SM-19}$$

33.1 Context

The paper writes: "Since $\xi = mH_0/kT\sqrt{2}$ is small (we defined here the Langevin parameter over the 'acting' magnetic field $H_0/\sqrt{2}$), using the expression $I/\tau_s = 6\eta\varphi$, one can eliminate the angle α from the solution (18). Up to the ξ^2 terms we have (19)." Two things must therefore be explained before we can derive anything: what the "acting field" means, and why the small parameter is ξ^2.

33.2 Symbols: the acting field and its Langevin parameter

Definition 33.2.1 (Acting field)**.** For a field oscillating as $H_0\cos\omega t$, the *acting* (root-mean-square) field is

$$H_{\text{act}} \equiv \sqrt{\overline{H^2}} = \sqrt{H_0^2\,\overline{\cos^2\omega t}} = \sqrt{H_0^2/2} = \frac{H_0}{\sqrt{2}}, \tag{33.2.1}$$

using the period average $\overline{\cos^2\omega t} = \tfrac{1}{2}$ derived as Eq. (27.4.5).

Throughout Sec. II of Shliomis & Morozov — and therefore throughout this volume and the next — the symbol ξ denotes the Langevin parameter formed with this acting field:

$$\boxed{\xi \equiv \frac{mH_0}{kT\sqrt{2}} \qquad\Longleftrightarrow\qquad \xi^2 = \frac{m^2H_0^2}{2(kT)^2}} \tag{33.2.2}$$

Remark 33.2.1 (A notational trap)*.* This is *not* the same ξ as in Sec. I of the paper, where $\xi = mH/kT$ was formed with the instantaneous field, nor the same as the $\xi_0 = mH_0/kT$ of Sec. III and of Bacri *et al.* The relation is $\xi_{\text{Sec. II}} = \xi_0/\sqrt{2}$. The authors chose it so that the results of Sec. II reduce, at $\omega = 0$, to the stationary formula (SM-3) with *no* stray factors — as we verify in Sec. 33.6. In this book we keep the paper's convention, flagging it wherever confusion is possible; the master notation table of Volume I records both.

33.3 Assumptions

1. Weak field: $\xi \ll 1$, and we work to order ξ^2 inclusive.

2. Everything from Chapter 32.

Why ξ^2 and not ξ? Because Θ, by (32.5.16), is proportional to MH_0, and M is itself proportional to χH_0. Two powers of the field are unavoidable — the same parity argument as for (SM-3) in Sec. 16.1. The first correction beyond leading order is therefore $O(\xi^2)$, and that is the order at which the paper works.

33.4 Step-by-Step Derivation

Step 1: solve for $\cos\alpha$ and $\sin\alpha$ in terms of $\tan\alpha$. For any angle with $\tan\alpha = x$ and $-\pi/2 < \alpha < \pi/2$,

$$\cos\alpha = \frac{1}{\sqrt{1+x^2}}, \qquad \sin\alpha = \frac{x}{\sqrt{1+x^2}}. \tag{33.4.1}$$

Derivation: from $\sin^2+\cos^2 = 1$, divide by $\cos^2\alpha$ to get $\tan^2\alpha + 1 = 1/\cos^2\alpha$, hence $\cos\alpha = (1+x^2)^{-1/2}$ (positive root, since α is in the first quadrant here), and then $\sin\alpha = \cos\alpha\tan\alpha = x/\sqrt{1+x^2}$.

Step 2: zeroth approximation. By (32.5.11), $\tan\alpha = (\omega - \Theta)\tau_B$. We do not yet know Θ; but (32.5.16) shows it is of order ξ^2 relative to ω (Step 4 below confirms this). To leading order, therefore,

$$\tan\alpha = \omega\tau_B + O(\xi^2). \tag{33.4.2}$$

Step 3: the magnitude M. Insert (33.4.2) into (33.4.1) and then into (32.5.6):

$$M = \chi H_0 \cos\alpha = \frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}, \tag{33.4.3}$$

the first relation of (SM-19). ✓ (The $O(\xi^2)$ correction to α would give an $O(\xi^2)$ correction to M, i.e. an $O(\xi^3)$ correction to the torque, beyond the order kept.)

Step 4: the spin Θ. From (32.5.16) with $\tau_s/I = 1/6\eta\varphi$:

$$\Theta = \frac{MH_0\sin\alpha}{6\eta\varphi} \tag{33.4.4}$$

$$= \frac{H_0}{6\eta\varphi}\cdot\frac{\chi H_0}{\sqrt{1+\omega^2\tau_B^2}}\cdot\frac{\omega\tau_B}{\sqrt{1+\omega^2\tau_B^2}} \qquad \text{using (33.4.3) and (33.4.1)} \tag{33.4.5}$$

$$= \frac{\chi H_0^2\,\omega\tau_B}{6\eta\varphi\big(1+\omega^2\tau_B^2\big)}. \tag{33.4.6}$$

Step 5: evaluate the prefactor. This is the step where all the physics of Volume II enters. We must simplify

$$\frac{\chi H_0^2 \tau_B}{6\eta\varphi}. \tag{33.4.7}$$

Insert $\chi = nm^2/3kT$ from (SM-15), $\tau_B = 3\eta V/kT$ from (SM-5), and $\varphi = nV$:

$$\frac{\chi H_0^2 \tau_B}{6\eta\varphi} = \frac{1}{6\eta(nV)} \cdot \frac{nm^2}{3kT} \cdot H_0^2 \cdot \frac{3\eta V}{kT} \tag{33.4.8}$$

$$= \frac{n\, m^2 H_0^2\, 3\eta V}{6\eta\, nV\, 3kT\, kT} \qquad \text{(collecting all factors)} \tag{33.4.9}$$

$$= \frac{m^2 H_0^2}{6(kT)^2} \qquad (n,\ V,\ \eta \text{ and the 3's cancel}) \tag{33.4.10}$$

$$= \frac{1}{6} \cdot \frac{m^2 H_0^2}{(kT)^2} \tag{33.4.11}$$

$$= \frac{1}{6} \cdot 2\xi^2 \qquad \text{by (33.2.2)} \tag{33.4.12}$$

$$= \frac{\xi^2}{3}. \tag{33.4.13}$$

This is the calculation that justifies the paper's definition of ξ over the acting field: with that definition, and only with it, the prefactor comes out as the clean $\xi^2/3$ printed in (SM-19).

Step 6: assemble Θ. Substituting (33.4.13) into (33.4.6):

$$\boxed{\Theta = \omega\, \frac{\xi^2/3}{1 + \omega^2\tau_B^2}} \tag{33.4.14}$$

the second relation of (SM-19). ✓ Note that this confirms, a posteriori, the ordering assumed in Step 2: $\Theta/\omega = O(\xi^2)$.

Step 7: the corrected phase. Now return to the exact relation (32.5.11) and insert (33.4.14):

$$\tan\alpha = \big(\omega - \Theta\big)\tau_B \tag{33.4.15}$$

$$= \omega\tau_B \left(1 - \frac{\Theta}{\omega}\right) \tag{33.4.16}$$

$$= \boxed{\omega\tau_B \left(1 - \frac{\xi^2/3}{1 + \omega^2\tau_B^2}\right)} \tag{33.4.17}$$

the third relation of (SM-19). ✓ ■

33.5 Consistency Checks

Zero frequency. $\omega = 0$: then $\alpha = 0$, $M = \chi H_0$ and $\Theta = 0$. A static field produces no lag, full magnetisation, and no spin — correct, since a permanent magnet does no work (Sec. 15.6).

Infinite frequency. $\omega\tau_B \to \infty$: $\alpha \to \pi/2$, $M \to \chi H_0/\omega\tau_B \to 0$, and $\Theta \to \omega\xi^2/(3\omega^2\tau_B^2) \to 0$. The moments cannot follow at all; the magnetisation dies and with it the spin. This is the behaviour the authors say "was expected" in their introduction — and it is why the negative-viscosity effect, which lives at $\omega\tau_B \gtrsim 1$ but not ∞, is a surprise.

Maximum spin. Where is Θ largest? Differentiate $\omega/(1+\omega^2\tau_B^2)$ with respect to ω:

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\frac{\omega}{1+\omega^2\tau_B^2} = \frac{\big(1+\omega^2\tau_B^2\big) - \omega\big(2\omega\tau_B^2\big)}{\big(1+\omega^2\tau_B^2\big)^2} = \frac{1-\omega^2\tau_B^2}{\big(1+\omega^2\tau_B^2\big)^2}, \tag{33.5.1}$$

which vanishes at $\omega\tau_B = 1$. The particles are dragged around fastest when the field frequency matches the Brownian rate. That the same combination $1-\omega^2\tau_B^2$ will appear in the numerator of the central result (SM-28) is not a coincidence; we shall meet it again in Volume V.

Dimensions. $[\chi H_0] = \mathrm{G}$ ✓; $[\Theta] = \mathrm{s}^{-1}$ since ξ and $\omega\tau_B$ are pure numbers and ω is s^{-1} ✓.

33.6 Physical Interpretation

Equation (33.4.3) is the classical Debye relaxation law: a system with a single relaxation time τ_B responds to a drive of frequency ω with an amplitude reduced by $(1+\omega^2\tau_B^2)^{-1/2}$ and a lag $\arctan\omega\tau_B$. It is the same formula that describes a dielectric, an RC circuit, or an overdamped oscillator. The ferrofluid is, magnetically, a Debye relaxator.

Equation (33.4.14) is more specific to our problem: it says the grains are dragged around at a rate that is only a fraction $\xi^2/3$ of the field frequency. In a weak field the moments are barely aligned, so the torque is weak, so the mechanical rotation is slow. Nevertheless it is *nonzero and steady*: a rotating field is a rotary motor for ferrofluid grains. That is the single most important qualitative fact of this chapter, because in the next two chapters the two counter-rotating motors are put into competition.

33.7 Connection to the Next Equation

Everything so far assumed a quiescent fluid. The two circular components are then exact mirror images, their spins cancel, and the linear field does nothing. Now we switch on the vorticity, which breaks the mirror symmetry. That is (SM-20).

Chapter 34

Equation (SM-20): The Vortex Breaks the Symmetry

$$
\begin{aligned}
M_+ &= \chi H_0 \cos\alpha_+, & M_- &= \chi H_0 \cos\alpha_-, \\
\Theta_+ &= (\tau_s/I) M_+ H_0 \sin\alpha_+ + \Omega, & \Theta_- &= (\tau_s/I) M_- H_0 \sin\alpha_- - \Omega, \\
\tan\alpha_+ &= (\omega - \Theta_+)\tau_B, & \tan\alpha_- &= (\omega - \Theta_-)\tau_B.
\end{aligned}
\tag{SM-20}
$$

34.1 Context

The paper's transition sentence is: "The degeneracy of the direction of rotation takes place only in the case of motionless fluid. Any flow with vorticity $\nabla \times \boldsymbol{v} = 2\boldsymbol{\Omega}$ ... eliminates this degeneracy because it destroys the symmetry between the 'plus' and 'minus' solutions. Let us show this."

34.2 Symbols

The subscripts $\pm$ now attach to *everything*: $M_\pm$, $\alpha_\pm$, $\Theta_\pm$ are the magnitude, lag and spin in the $+$ and $-$ circular fields respectively, and they are no longer equal. $\boldsymbol{\Omega} = (0, 0, \Omega)$ is the imposed hydrodynamic vorticity.

34.3 Assumptions

1. As before: weak field, steady state, rigid dipoles.
2. **The vorticity is along** z, i.e. perpendicular to the plane in which the field rotates. This is $\beta = \pi/2$ again, and it is the experimental geometry (Sec. 3.5).
3. The ansatz (SM-17) survives, with different constants for the two components. This requires the vorticity to be steady and uniform on the scale of a grain, which it is.

34.4 Step-by-Step Derivation

Step 1: what changes. The magnetisation equation (SM-7) is unchanged in form; it involves the particle spin $\boldsymbol{\Theta}$, which is still along z. The only equation that changes is (SM-11), which

now has $\boldsymbol{\Omega} \neq 0$:

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{34.4.1}$$

Step 2: the + field. The ansatz is $\boldsymbol{M}_+ = M_+ \hat{\boldsymbol{e}}_r^{(+)}$ with $\hat{\boldsymbol{e}}_r^{(+)}$ at angle $\omega t - \alpha_+$, and $\boldsymbol{\Theta}_+ = \Theta_+ \hat{\boldsymbol{z}}$. Repeating Steps 1–7 of Sec. 32.5 verbatim — nothing in them used the value of $\boldsymbol{\Theta}$, only its direction — gives the same two relations with subscripts attached:

$$M_+ = \chi H_0 \cos\alpha_+, \tag{34.4.2}$$

$$\tan\alpha_+ = \big(\omega - \Theta_+\big)\tau_B. \tag{34.4.3}$$

Step 8, however, now reads

$$\boldsymbol{\Theta}_+ = \boldsymbol{\Omega} + \frac{\tau_s}{I} M_+ H_0 \sin\alpha_+ \, \hat{\boldsymbol{z}}, \tag{34.4.4}$$

whose z component is

$$\Theta_+ = \frac{\tau_s}{I} M_+ H_0 \sin\alpha_+ + \Omega. \tag{34.4.5}$$

Equations (34.4.2)–(34.4.5) are the left column of (SM-20). ✓

Step 3: the − field, and a careful word about signs. For the − field the magnetisation rotates clockwise and the spin points along $-z$. Following the paper we write the ansatz with explicit signs, as in (SM-17):

$$\boldsymbol{M}_- = \big[M_- \cos(\omega t - \alpha_-),\ -M_- \sin(\omega t - \alpha_-),\ 0\big], \qquad \boldsymbol{\Theta}_- = \big(0, 0, -\Theta_-\big), \tag{34.4.6}$$

so that M_-, α_- and Θ_- are all *positive magnitudes*. The mirror operation $y \to -y$ maps the + problem onto the − problem *except for the vorticity*, which is a pseudo-vector along z and does *not* change sign under this mirror. Concretely, applying the mirror to the + results (34.4.2)–(34.4.5) and remembering that $\boldsymbol{\Omega}$ keeps its sign:

$$M_- = \chi H_0 \cos\alpha_-, \tag{34.4.7}$$

$$-\Theta_- = -\frac{\tau_s}{I} M_- H_0 \sin\alpha_- + \Omega \qquad \Longrightarrow \qquad \Theta_- = \frac{\tau_s}{I} M_- H_0 \sin\alpha_- - \Omega, \tag{34.4.8}$$

$$\tan\alpha_- = \big(\omega - \Theta_-\big)\tau_B. \tag{34.4.9}$$

These are the right column of (SM-20). ✓ ■

The asymmetry, isolated. Compare (34.4.5) and (34.4.8): the vorticity enters with a + sign in one and a − sign in the other. *That single sign difference is the entire origin of the effect.* Everything that follows in this volume and the next is bookkeeping applied to it.

34.5 Physical Interpretation

The authors put it in one sentence: "It can be seen that the rotation of the particles is faster when the field and the fluid are rotating in the same direction."

Consider a grain sitting in a shear flow that turns it counterclockwise at rate Ω. Now apply the + (counterclockwise) circular field. Field and flow cooperate: the grain turns at $\Theta_+ = \Omega +$ (magnetic drive). Apply instead the − field: they oppose, and the grain turns clockwise at Θ_- where the magnetic drive has been *reduced* by Ω.

Now recall that the linearly polarised field is the sum of the two. In a quiescent fluid the two drives were equal and cancelled. With a vortex they no longer cancel, and the surviving imbalance is what we are after. Whether that imbalance helps or hinders the flow — whether the grains end up turning slower or faster than the liquid — is not yet obvious from (SM-20); it takes until Volume V to settle. The answer will turn out to depend on frequency, and to change sign at $\omega\tau_B = 1$.

34.6 Connection to the Next Equation

Equations (SM-20) are still implicit: $\alpha_\pm$ appears on both sides. The paper now simplifies them by keeping only the leading terms, giving (SM-21).

Chapter 35

Equation (SM-21): The Principal Terms

$$\tan\alpha_+ = \big(\omega - \Omega\big)\tau_B, \qquad \tan\alpha_- = \big(\omega + \Omega\big)\tau_B. \tag{SM-21}$$

35.1 Context

"Retaining only the principal terms in the expression for $\tan\alpha_\pm$ in (20), we have (21)." We must say precisely which terms are principal and why the others may be dropped.

35.2 Assumptions

Assumption 35.2.1 (The magnetic drive is negligible compared with the vorticity in the phase relation)**.** In $\Theta_\pm = \pm\Omega +$ (magnetic drive), the magnetic drive is of order $\omega\xi^2/3$ by (33.4.14), and it is dropped from the expression for $\tan\alpha_\pm$ — but not the Ω.

Justification. Two separate points must be made.

(i) Why the magnetic term may be dropped. We are computing, ultimately, a quantity of order ξ^2 (the rotational viscosity) linear in Ω. Keeping the magnetic contribution to $\Theta_\pm$ inside $\tan\alpha_\pm$ would generate corrections of relative order ξ^2, i.e. terms of order $\xi^4\Omega$ in the final answer — beyond the accuracy of the weak-field theory.

(ii) Why the Ω may not *be dropped, even though $\Omega\tau_B \ll 1$.* Because it is the only thing that distinguishes the two components. Setting $\Omega = 0$ here would make $\alpha_+ = \alpha_-$, the two solutions would again be mirror images, and the entire effect would vanish identically. We are computing a quantity that is *first order* in Ω and we must therefore keep Ω to first order everywhere, however small it is. This is the standard structure of linear-response theory: the small parameter is retained to exactly the order of the response being computed.

35.3 Step-by-Step Derivation

Step 1. From (34.4.3) and (34.4.5):

$$\tan\alpha_+ = \big(\omega - \Theta_+\big)\tau_B \tag{35.3.1}$$

$$= \Big(\omega - \Omega - \underbrace{\tfrac{\tau_s}{I} M_+ H_0 \sin\alpha_+}_{O(\omega\xi^2)}\Big)\tau_B \tag{35.3.2}$$

$$= \big(\omega - \Omega\big)\tau_B + O(\xi^2). \tag{35.3.3}$$

Step 2. From (34.4.9) and (34.4.8):

$$\tan\alpha_- = \big(\omega - \Theta_-\big)\tau_B \tag{35.3.4}$$

$$= \Big(\omega + \Omega - \underbrace{\tfrac{\tau_s}{I} M_- H_0 \sin\alpha_-}_{O(\omega\xi^2)}\Big)\tau_B \tag{35.3.5}$$

$$= \big(\omega + \Omega\big)\tau_B + O(\xi^2). \tag{35.3.6}$$

Dropping the indicated corrections gives (SM-21). ■

35.4 Physical Interpretation

The result has a one-line reading which is worth more than the algebra: *a grain carried round by the flow sees the two circular fields at different frequencies.* In the frame of the rotating grain, the co-rotating field appears slowed to $\omega - \Omega$, and the counter-rotating one appears speeded to $\omega + \Omega$. This is a Doppler shift for rotations. Each component then behaves as a Debye relaxator at *its own* apparent frequency, and since the Debye response is a nonlinear function of frequency, the two responses no longer cancel.

That is the mechanism in its purest form, and everything remaining is the computation of how much they fail to cancel.

35.5 Connection to the Next Equation

With the two phase angles known, we can compute the transverse magnetisation generated by each component. That is (SM-22).

Chapter 36

Equation (SM-22): The Transverse Magnetisation of Each Component

$$M_y^{(+)} = \chi H_0 \cos\alpha_+ \sin(\omega t - \alpha_+) = \frac{\chi H_0}{1 + (\omega - \Omega)^2 \tau_B^2} \Big[\sin\omega t - (\omega - \Omega)\tau_B \cos\omega t \Big],$$
$$M_y^{(-)} = -\chi H_0 \cos\alpha_- \sin(\omega t - \alpha_-) = -\frac{\chi H_0}{1 + (\omega + \Omega)^2 \tau_B^2} \Big[\sin\omega t - (\omega + \Omega)\tau_B \cos\omega t \Big]. \quad \text{(SM-22)}$$

36.1 Context

"Owing to the difference between the angles α_+ and α_-, the superposition of the fields $\boldsymbol{H}_+$ and $\boldsymbol{H}_-$ generates the nonzero y component of magnetization, contrary to the case of quiescent fluid." The y component is the object we want: it is perpendicular to the applied linear field (which lies along x), and a magnetisation perpendicular to the field is exactly what produces a torque.

36.2 Symbols

$M_y^{(\pm)}$ is the y component of the magnetisation produced by the $\pm$ circular field *acting alone*, i.e. before the superposition of Chapter 37. Note that these are not halved yet: the factor $\frac{1}{2}$ enters in (SM-23).

36.3 Mathematical Background: a useful trigonometric reduction

Result 36.3.1. *For any angle α with $\tan\alpha = x$,*

$$\cos\alpha \, \sin(\omega t - \alpha) = \frac{\sin\omega t - x\cos\omega t}{1 + x^2}. \tag{36.3.1}$$

Derivation. Expand the sine of a difference, $\sin(A - B) = \sin A\cos B - \cos A\sin B$, with $A = \omega t$, $B = \alpha$:

$$\cos\alpha \, \sin(\omega t - \alpha) = \cos\alpha \Big[\sin\omega t \cos\alpha - \cos\omega t \sin\alpha \Big] \tag{36.3.2}$$

$$= \cos^2\alpha\,\sin\omega t - \cos\alpha\sin\alpha\,\cos\omega t. \tag{36.3.3}$$

Now use (33.4.1):

$$\cos^2\alpha = \frac{1}{1+x^2}, \tag{36.3.4}$$

$$\cos\alpha\sin\alpha = \frac{1}{\sqrt{1+x^2}}\cdot\frac{x}{\sqrt{1+x^2}} = \frac{x}{1+x^2}. \tag{36.3.5}$$

Substituting into (36.3.3):

$$\cos\alpha\sin(\omega t-\alpha) = \frac{\sin\omega t}{1+x^2} - \frac{x\cos\omega t}{1+x^2} = \frac{\sin\omega t - x\cos\omega t}{1+x^2}. \quad \blacksquare \tag{36.3.6}$$

36.4 Step-by-Step Derivation

Step 1: read off M_y from the ansatz. By (SM-17), the y component of the magnetisation in the $\pm$ field is

$$M_y^{(\pm)} = \pm M_\pm \sin(\omega t - \alpha_\pm). \tag{36.4.1}$$

Step 2: insert the magnitudes. By (34.4.2) and (34.4.7), $M_\pm = \chi H_0\cos\alpha_\pm$, so

$$M_y^{(\pm)} = \pm\chi H_0\cos\alpha_\pm\sin(\omega t-\alpha_\pm), \tag{36.4.2}$$

which is the middle expression of each line of (SM-22). ✓

Step 3: apply the reduction. Use (36.3.1) with $x = x_\pm \equiv \tan\alpha_\pm$, and then (SM-21), which gives $x_+ = (\omega-\Omega)\tau_B$ and $x_- = (\omega+\Omega)\tau_B$:

$$M_y^{(+)} = \chi H_0\,\frac{\sin\omega t - (\omega-\Omega)\tau_B\cos\omega t}{1+(\omega-\Omega)^2\tau_B^2}, \tag{36.4.3}$$

$$M_y^{(-)} = -\chi H_0\,\frac{\sin\omega t - (\omega+\Omega)\tau_B\cos\omega t}{1+(\omega+\Omega)^2\tau_B^2}. \tag{36.4.4}$$

These are exactly the right-hand expressions of (SM-22). ■

36.5 Physical Interpretation

Each line of (36.4.3)–(36.4.4) contains two pieces:

- a term $\propto \sin\omega t$, which is *in quadrature* with the driving field $H_0\cos\omega t$; and
- a term $\propto \cos\omega t$, which is *in phase* with it.

In the language of linear response these are the dissipative and reactive parts. The essential structure to notice is that the two components differ *only* through the replacement $\omega \to \omega \mp \Omega$ — the rotational Doppler shift of Sec. 35.4. If Ω were zero the two lines would be identical apart from the overall minus sign, and their sum would be zero: no transverse magnetisation, no torque, nothing.

36.6 Connection to the Next Equation

Adding the two and expanding to first order in Ω gives the central result of this volume, (SM-23).

Chapter 37

Equations (SM-23) and (SM-24): The Transverse Magnetisation

$$M_y = \tfrac{1}{2}\big(M_y^{(+)} + M_y^{(-)}\big) = \Omega\tau_B\,\chi H_0 \cos^2\alpha\,\cos(\omega t - 2\alpha), \tag{SM-23}$$

$$\tan\alpha = \omega\tau_B. \tag{SM-24}$$

37.1 Context

This is the destination of the whole circular-polarisation construction. The paper prefaces it with the observation that "the Brownian rotational diffusion time for the particles with diameter 10^{-6} cm is usually not higher than 10^{-5} s. Therefore the inequality $\Omega\tau_B \ll 1$ is almost always valid", which is the licence for the linearisation performed below.

37.2 Symbols

M_y is the y component of the magnetisation in the *linearly polarised* field, i.e. the physically relevant quantity. α is now the phase lag evaluated at the *unshifted* frequency, defined by (SM-24). Note that this α is the α of (SM-19) with the ξ^2 correction discarded — a point the authors flag and which we discuss in Sec. 37.6.

37.3 Assumptions

Assumption 37.3.1 (Slow flow)**.** $\Omega\tau_B \ll 1$, so that M_y may be computed to first order in Ω.

Numerically, Bacri *et al.* have $\Omega < 60\ \mathrm{s}^{-1}$ and $\tau_B = 1.6$ ms, so $\Omega\tau_B < 0.1$ (Sec. 3.5). ✓

37.4 Mathematical Background: linearisation of a function of a shifted argument

We must add two expressions that differ only by $\omega \to \omega \mp \Omega$. Define

$$F(x) \equiv \frac{\sin\omega t - x\cos\omega t}{1 + x^2}, \tag{37.4.1}$$

so that, by (36.4.3)–(36.4.4), $M_y^{(+)} = \chi H_0 F(x_+)$ and $M_y^{(-)} = -\chi H_0 F(x_-)$ with $x_\pm = (\omega \mp \Omega)\tau_B$. Writing $x_0 \equiv \omega\tau_B$ and $\delta \equiv \Omega\tau_B$, we have $x_\pm = x_0 \mp \delta$ and therefore need

$$F(x_0 - \delta) - F(x_0 + \delta). \tag{37.4.2}$$

Taylor-expand each term about x_0 — legitimate because F is smooth and $\delta \ll 1$ by Assumption 37.3.1:

$$F(x_0 - \delta) = F(x_0) - \delta F'(x_0) + \tfrac{1}{2}\delta^2 F''(x_0) - \cdots, \tag{37.4.3}$$
$$F(x_0 + \delta) = F(x_0) + \delta F'(x_0) + \tfrac{1}{2}\delta^2 F''(x_0) + \cdots. \tag{37.4.4}$$

Subtracting, the even-order terms cancel identically and

$$F(x_0 - \delta) - F(x_0 + \delta) = -2\delta F'(x_0) + O(\delta^3). \tag{37.4.5}$$

Note the structure: the difference is *odd* in δ, hence odd in Ω — as it must be, since reversing the sense of the shear must reverse the sense of the transverse magnetisation.

The derivative. Differentiate (37.4.1) by the quotient rule, $(u/v)' = (u'v - uv')/v^2$, with $u = \sin\omega t - x\cos\omega t$ (so $u' = -\cos\omega t$, the time t being held fixed) and $v = 1 + x^2$ (so $v' = 2x$):

$$F'(x) = \frac{-\cos\omega t\big(1 + x^2\big) - \big(\sin\omega t - x\cos\omega t\big)2x}{\big(1 + x^2\big)^2} \tag{37.4.6}$$
$$= \frac{-\cos\omega t - x^2\cos\omega t - 2x\sin\omega t + 2x^2\cos\omega t}{\big(1 + x^2\big)^2} \tag{37.4.7}$$
$$= \frac{\big(x^2 - 1\big)\cos\omega t - 2x\sin\omega t}{\big(1 + x^2\big)^2}. \tag{37.4.8}$$

37.5 Step-by-Step Derivation of (SM-23)

Step 1: assemble the sum.

$$M_y = \tfrac{1}{2}\big(M_y^{(+)} + M_y^{(-)}\big) \tag{37.5.1}$$
$$= \tfrac{1}{2}\chi H_0\Big[F(x_+) - F(x_-)\Big] \tag{37.5.2}$$
$$= \tfrac{1}{2}\chi H_0\Big[F(x_0 - \delta) - F(x_0 + \delta)\Big] \tag{37.5.3}$$
$$= \tfrac{1}{2}\chi H_0\big(-2\delta\big)F'(x_0) \qquad \text{by (37.4.5)} \tag{37.5.4}$$
$$= -\chi H_0\,\Omega\tau_B\,F'(x_0). \tag{37.5.5}$$

Step 2: insert the derivative. With (37.4.8) at $x = x_0$:

$$M_y = -\chi H_0 \Omega\tau_B\,\frac{\big(x_0^2 - 1\big)\cos\omega t - 2x_0\sin\omega t}{\big(1 + x_0^2\big)^2} \tag{37.5.6}$$
$$= \chi H_0 \Omega\tau_B\,\frac{\big(1 - x_0^2\big)\cos\omega t + 2x_0\sin\omega t}{\big(1 + x_0^2\big)^2}, \tag{37.5.7}$$

the overall minus sign having been absorbed by reversing both terms in the numerator.

Step 3: recognise a double angle. Define α by $\tan\alpha = x_0 = \omega\tau_B$, which is Eq. (SM-24). The double-angle formulas in terms of the tangent are

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = \frac{1}{1+x_0^2} - \frac{x_0^2}{1+x_0^2} = \frac{1-x_0^2}{1+x_0^2}, \tag{37.5.8}$$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha = 2\cdot\frac{x_0}{\sqrt{1+x_0^2}}\cdot\frac{1}{\sqrt{1+x_0^2}} = \frac{2x_0}{1+x_0^2}, \tag{37.5.9}$$

both obtained from (33.4.1). Therefore

$$1 - x_0^2 = \left(1+x_0^2\right)\cos 2\alpha, \qquad 2x_0 = \left(1+x_0^2\right)\sin 2\alpha. \tag{37.5.10}$$

Step 4: substitute. Insert (37.5.10) into (37.5.7):

$$M_y = \chi H_0 \Omega\tau_B \,\frac{\left(1+x_0^2\right)\left[\cos 2\alpha\cos\omega t + \sin 2\alpha\sin\omega t\right]}{\left(1+x_0^2\right)^2} \tag{37.5.11}$$

$$= \chi H_0 \Omega\tau_B \,\frac{\cos 2\alpha\cos\omega t + \sin 2\alpha\sin\omega t}{1+x_0^2}. \tag{37.5.12}$$

Step 5: collapse the bracket. The addition formula $\cos(A-B) = \cos A\cos B + \sin A\sin B$ with $A = \omega t$ and $B = 2\alpha$ turns the numerator into $\cos(\omega t - 2\alpha)$. And by (33.4.1), $1/(1+x_0^2) = \cos^2\alpha$. Hence

$$\boxed{M_y = \Omega\tau_B\,\chi H_0\,\cos^2\alpha\,\cos\left(\omega t - 2\alpha\right)} \tag{SM-23}$$

which is Eq. (SM-23) exactly. ■

37.6 Why (SM-24) Discards the ξ^2 Term

The paper is careful here, and so must we. In (SM-19) the phase was

$$\tan\alpha = \omega\tau_B\left(1 - \frac{\xi^2/3}{1+\omega^2\tau_B^2}\right), \tag{37.6.1}$$

yet (SM-24) uses simply $\tan\alpha = \omega\tau_B$. The authors explain: "which is obtained from (19) after discarding the ξ^2 term. The last procedure is necessary because in the presence of nonlinear terms the half of the sum $\boldsymbol{M}_+ + \boldsymbol{M}_-$ is not equal to the magnetization in the linearly polarized field. The principle of superposition is valid only within the framework of the linear response."

Unpacked, the argument is this. The step $\boldsymbol{M} = \frac{1}{2}(\boldsymbol{M}_+ + \boldsymbol{M}_-)$ used in Step 1 above is *only* valid if the magnetisation responds linearly to the field. The ξ^2 term in (SM-19) is precisely a signature of nonlinearity: it arises from the back-reaction of the particle spin, which is quadratic in the field. Retaining it while simultaneously using superposition would be inconsistent — it would keep a nonlinear correction in a framework that assumes linearity. The consistent procedure at this order is to use the strictly linear response, $\tan\alpha = \omega\tau_B$, throughout. Any ξ^2 refinements belong to the finite-amplitude theory of Sec. III of the paper, which abandons superposition altogether and which we shall reach in Volume VI.

37.7 Physical Interpretation

Equation (SM-23) is the pivot of the whole subject. Read it slowly.

(i) $M_y \propto \Omega$**.** No shear, no transverse magnetisation. In a quiescent fluid the linearly polarised field produces a magnetisation strictly along x, hence no torque, hence nothing. The vorticity is not a small perturbation to the effect — it is its cause.

(ii) $M_y \propto \tau_B$**.** No memory, no transverse magnetisation either. If the moments could follow the field instantaneously they would always be along x regardless of the flow.

(iii) $M_y \propto \chi H_0$**.** One power of the field, as expected for a linear response. The second power will arrive when this magnetisation is crossed with the field to make a torque, restoring the H_0^2 (i.e. ξ^2) scaling of the final answer.

(iv) The phase 2α **is the crux.** The transverse magnetisation oscillates as $\cos(\omega t - 2\alpha)$, i.e. lagging the field $H_0 \cos \omega t$ by *twice* the single-component lag. Why twice? Because M_y arises from the *difference* between two responses whose phases are shifted in opposite directions — effectively a derivative of the response with respect to frequency, which doubles the phase. This factor of two is the entire reason the effect can change sign, as follows.

The torque will be built from the product $M_y H_x \propto \cos(\omega t - 2\alpha) \cos \omega t$, whose time average is (by the addition formula and the period averages of Sec. 27.4)

$$\overline{\cos(\omega t - 2\alpha)\cos \omega t} = \overline{\cos 2\alpha \cos^2 \omega t + \sin 2\alpha \sin \omega t \cos \omega t} = \tfrac{1}{2} \cos 2\alpha. \tag{37.7.1}$$

And $\cos 2\alpha = (1 - \omega^2 \tau_B^2)/(1 + \omega^2 \tau_B^2)$ by (37.5.8) **changes sign at** $\omega \tau_B = 1$**.** A single lag α never exceeds $\pi/2$ and $\cos \alpha$ is always positive; but 2α passes through $\pi/2$ when $\alpha = \pi/4$, i.e. when $\omega \tau_B = 1$, and beyond that point the average torque reverses.

That is the negative-viscosity effect. Everything in Volume V is the careful execution of the calculation previewed in (37.7.1).

37.8 Connection to the Next Equation

Having a transverse magnetisation, the paper computes the torque it exerts on the fluid. That is (SM-25), the first equation of Volume V.

Chapter 38

Status and Road Map

38.1 What has been established in Volume IV

Equation	Statement	Where derived
(SM-16)	$\boldsymbol{H}_\pm = (H_0\cos\omega t, \pm H_0\sin\omega t, 0)$, $\boldsymbol{H} = \frac{1}{2}(\boldsymbol{H}_+ + \boldsymbol{H}_-)$	Ch. 30
(SM-17)	rotating ansatz with lag α and axial spin $\pm\Theta$, justified by symmetry	Ch. 31
(SM-18)	$M = \chi H_0\cos\alpha$; $\Theta = (\tau_s/I)MH_0\sin\alpha$; $\tan\alpha = (\omega-\Theta)\tau_B$	Ch. 32
(33.2.2)	the "acting field" convention $\xi = mH_0/kT\sqrt{2}$	Sec. 33.2
(SM-19)	$M = \chi H_0/\sqrt{1+\omega^2\tau_B^2}$; $\Theta = \omega(\xi^2/3)/(1+\omega^2\tau_B^2)$; corrected $\tan\alpha$	Ch. 33
(SM-20)	the same relations with $\pm\Omega$ added to the spins	Ch. 34
(SM-21)	$\tan\alpha_\pm = (\omega\mp\Omega)\tau_B$: the rotational Doppler shift	Ch. 35
(SM-22)	$M_y^{(\pm)}$ for each circular component	Ch. 36
(SM-23)	$M_y = \Omega\tau_B\chi H_0\cos^2\alpha\cos(\omega t - 2\alpha)$	Ch. 37
(SM-24)	$\tan\alpha = \omega\tau_B$, and why the ξ^2 term must be dropped	Sec. 37.6

38.2 Cumulative progress

Shliomis & Morozov: Eqs. (SM-1)–(SM-24) complete. Bacri *et al.*: Eqs. (B-1)–(B-4) and (B-10) complete. Remaining:

- **Volume V** — Eqs. (SM-25)–(SM-34): the instantaneous torque $\boldsymbol{M}\times\boldsymbol{H}$; the separation of fast and slow time scales and the averaged momentum equation (SM-26); the rotational viscosity (SM-27); the central result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2} \tag{SM-28}$$

  with its sign change at $\omega_0\tau_B = 1$, its minimum at $\omega_m\tau_B = \sqrt{3}$ and the value $-\Delta\eta(0)/8$ there; the high-frequency asymptote (SM-29); the second, direct variant of the solution

(SM-30)–(SM-32); and the formula for the mean particle spin (SM-33) leading back to (SM-34).

- **Volume VI** — Eqs. (SM-35)–(SM-43): arbitrary field amplitude.
- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method and the comparison with experiment.

End of Volume IV. Volume V begins with Eq. (SM-25).